University
Mohammed VI
Polytechnic

College of Computing

Lab 5 : Normalization

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Team Name	The Avengers
Member 1	Aouatif Guennoun
Member 2	Ranya El Bahali
Member 3	Hayat Ennani
Member 4	Valdez Kemekong
Repository Link	`https://github.com/aouatif-Gu/da`

1 Introduction

This deliverable covers database normalization, identifying candidate keys and functional dependencies, checking BCNF, and decomposing relations while ensuring losslessness and dependency preservation.

2 Requirements

- Identify relations that violate BCNF and justify why.
- Determine candidate keys and functional dependencies.
- Identify BCNF violations and the strongest normal form satisfied.
- Decompose relations into BCNF where necessary.
- Verify lossless decomposition and dependency preservation.
- Document team roles, challenges, solutions, design choices, and assumptions.

3 Methodology

The role of each team member :

Valdez Kemekong : shared his ideas with the team, helped figuring out the solutions and explained to the team members what was ambiguous in the lecture .

Aouatif Guennoun : was in charge of writing down the eventual solutions and discussed with the team the possible ways of solving the tasks.

Ranya El Bahali : was in charge of writing the repor, shared ideas with the team members and gave examples to check and test our answers .

Hayat Ennani :was in charge of writing down the report, gave her opinion about the eventual solutions and discussed the correctness of the team's suggestions for the solution.

4 Implementation & Results

1. List any relations in the Lab 4 Solution that are not BCNF. Justify.

There are no FDs that are not BCNF in the last lab's solution, because all of the FDs that were not trivial FDs their determinants were superkey which does not violate BCNF.

a) **Equipment Inventory**

Candidate key: serialNo
The relation violates the BCNF
The FD responsible for the violation : modelId $\rightarrow$ manufacturer
Justification :This FD is not trivial and modelId is not a superkey because it doesn't determine the whole relation : $modelId^{+}$={manufacturer,modelId}
The strongest normal form it satisfies is : 1NF.
The decomposition: R1(modelId,manifacturer) and R2(serialNo, modelId, labId)

The decomposition is lossless because $R1 \cap R2$ is the attribute `modelId` which is a key of $R1$.
All the relevant FDs are preserved because we can extract them all from the two relations $R2$ and $R1$ without joining them back together.

b) **Project Participation**:

Candidate key: (pid,projectCode)
The relation violates the BCNF
The FD responsible for the violation : projectCode $\rightarrow$ piPid
Justification :This FD is not trivial and projectCode is not a superkey because it doesn't determine the whole relation : $projectCode^{+}$={piPid,projectCode
The strongest normal form it satisfies is : 1NF.
Justification : piPid is not a part of some candidate key and the FD already violates BCNF so it violates 3NF. And since piPid depends only on projectCode then this FD violates the 2NF as well that's why the strongest NF it satisfies is 1NF.
The decomposition: R1(projectCode, piPid) and R2(pid, projectCode, role)
The decomposition is lossless because $R1 \cap R2$ is the attribute `projectCode` which is a key of $R1$.
All the relevant FDs are preserved because we can extract them all from the two relations $R2$ and $R1$ without joining them back together.

c) Equipment Certification Requirement:

Candidate key: (modelId, certCode)
The relation violates the BCNF
The FD responsible for the violation is :
certCode $\rightarrow$ certTitle, issuingAthority
The strongest NF is : 1NF
Justification: This FD is not trivial and certCode is not a super key because: $certCode^{+}$={certTitle, issuingAuthority, certCode}, so the FD violates BCNF, since certTitle, issuingAthority is not part of the candidate key (modelId, certCode) this FD violates 3NF as well.
The decomposition: R1(modelId, certCode) and R2(certCode, certTitle, issuingAuthority)
The decomposition is Lossless because $R1 \cap R2$ is the attribute `certCode` which is a key of $R2$
All the relevant FDs are preserved because we can extract them all from the two relations $R2$ and $R1$ without joining them back together.

d) Supplier Office:

The Candidate Key: (suppId, consId, labId)
The relation violates the BCNF
The FD responsible for the violation is :
suppId $\rightarrow$ supplierName
The strongest NF is : 1NF
Justification: This FD is not trivial and suppId is not a super key because: $suppId^{+}$={suppId,

supplierName}, so the FD violates BCNF, and since supplierName is not part of the candidate key (suppId, consId, labId) thus this FD violates 3NF as well.
The decomposition: R1(suppId, supplierName) and R2(suppId, consId, labId, unitPrice)
The decomposition is Lossless because $R1 \cap R2$ is the attribute `suppId` which is a key of $R1$
All the relevant FDs are preserved because we can extract them all from the two relations $R2$ and $R1$ without joining them back together.

e) Reservation and Project Information :
The Candidate Key: resId
The relation violates BCNF
The FD responsible for this violation is :
projectCode → projectTitle
The strongest NF is : 1NF
Justification: This FD is not trivial and projectCode is not a super key because:
$projectCode^{+}$={projectCode,projectTitle}, so the FD violates BCNF, and since projectTitle is not part of the candidate key resId thus this FD violates 3NF as well.
The decomposition: R1(projectCode,projectTitle) and R2(resId, projectCode, madeByP id, status)
The decomposition is Lossless because $R1 \cap R2$ is the attribute `projectCode` which is a key of $R1$
All the relevant FDs are preserved because we can extract them all from the two relations $R2$ and $R1$ without joining them back together.

5 Discussion

One of the main challenges that we faced during this lab is that for us to decompose the relation that violates the BCNF we had two ways of doing it : the first way is decomposing according to the key of the main relation which gives a lossless decompostion but in some cases we loose some FDs, that's why decomposing according to the left attribute of the FD that violates the BNCF was the solution .

6 Conclusion

The deliverable analyzes BCNF violations in relational schemas, identifies candidate keys and functional dependencies, and decomposes relations where necessary. It evaluates losslessness and dependency preservation to improve database design and reduce redundancy.